HireX — Quality Assurance Test Plan

Document ID: `HXP-HANDOVER-01`
Platform Version: HireX V0 / Enterprise Multi-Tenant Recruitment Suite
Target Audience: QA Lead, Manual & Automation Test Engineers, Engineering Leadership
Last Updated: October 2026

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1. Objectives & Scope

1.1 Objectives

The objective of this Test Plan is to define the testing strategy, scope, environment configurations, execution procedures, and pass/fail criteria to guarantee that HireX operates reliably across all candidate, recruiter, interviewer, and backend service boundaries.

1.2 In-Scope Functionality

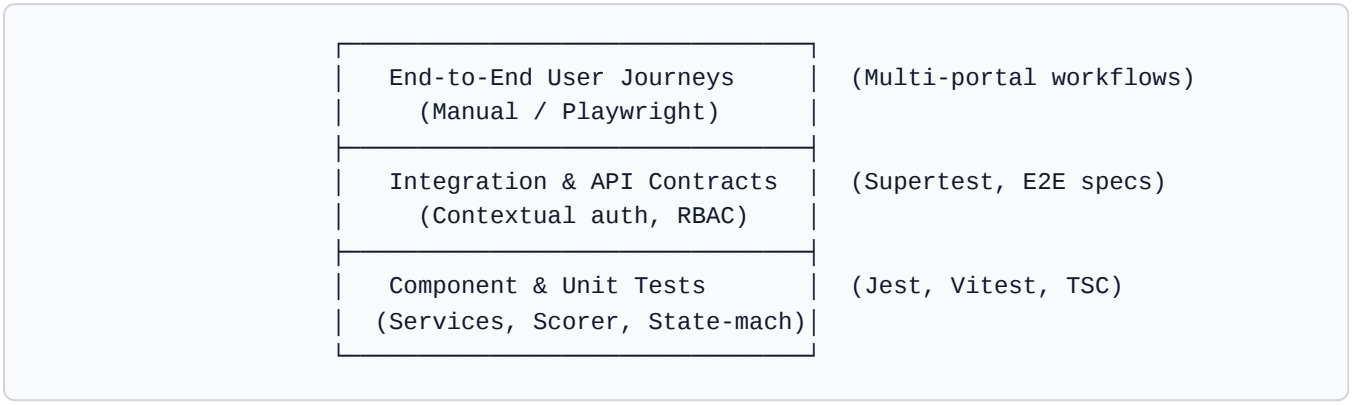
- **Authentication & Authorization:** Credential signup/login, invite token onboarding, session cookie partitioning (ADR-001), role-based access control (RBAC), and password reset flows.
- **Candidate Portal:** Public/branded job discovery, dynamic multi-step application form wizard with local autosave drafts, interview slot self-booking, offer review/signing, and onboarding document uploads.
- **Recruiter / Employer Dashboard:** Multi-tenant company onboarding, job requisition with custom form builder, Kanban pipeline state machine, AI resume scoring (`aiScore`), interview scheduling, PDF offer generation, and document verification.
- **Interviewer Workspace:** Upcoming interview queue, recurring weekly availability calendar, candidate resume review, private recruiter briefings, and structured scorecards with Cultural Fit rating.
- **Integrations & Webhooks:** Schedular (MeetX) video meeting creation, HMAC-signed webhooks (`meeting.ended`, `meeting.mom.ready`), Cloudflare R2 file storage, and XML job distribution feeds.
- **Cross-Cutting Concerns:** Forward-only state machine integrity, concurrency slot booking guards, and tenant data isolation.

1.3 Out-of-Scope (Limitations for Current Stage)

- Direct third-party posting APIs to live LinkedIn/Naukri (fail-closed fallbacks and XML feeds are tested; live vendor partner credentials not yet active).
- Real WebRTC audio/video packet quality measurements under 2G network throttles (out of scope for web application testing).
- Real payment processing or payroll integrations (not present in HireX V0).

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2. Test Strategy & Test Levels



1. **Unit Testing:** Focuses on business rules, calculation utilities (e.g. `ai-match-scorer.ts`, `resume-text.ts`), DTO validation, and service isolation.
2. **Integration / API Testing:** Executes HTTP requests against the NestJS testing module with in-memory or PostgreSQL test databases, verifying session resolution, tenant isolation, and status guards.
3. **End-to-End (E2E) Journey Testing:** Validates complete multi-actor flows across browser contexts (Recruiter publishes job -> Candidate applies -> Interviewer evaluates -> Candidate hired).
4. **Security & Session Isolation Testing:** Validates cookie boundaries, route guards, token expirations, and cross-portal access denial.

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3. Test Environments & Prerequisites

3.1 Environment Topology

Environment	Portal / Service	URL	Notes
Local Dev	Backend API	`http://localhost:3001`	PostgreSQL local DB, `.env` configured
Local Dev	Candidate Portal	`http://localhost:3000`	Next.js App Router
Local Dev	Recruiter Portal	`http://localhost:3002`	Next.js App Router
Local Dev	Interviewer Portal	`http://localhost:3003`	Next.js App Router
Hosted Staging	Backend API	`https://api.dev.hire.zobay.in`	Linux container, Cloudflare R2 connected

Environment	Portal / Service	URL	Notes
Hosted Staging	Candidate Portal	`https://dev.candidate.hire.zobay.in`	CNAME reverse proxy active
Hosted Staging	Recruiter Portal	`https://dev.recruiter.hire.zobay.in`	Org administration & pipeline
Hosted Staging	Interviewer Portal	`https://dev.interviewer.hire.zobay.in`	Technical panel workspace

3.2 Pre-Configured Test Accounts

Role	Portal Login	Test Email	Password	Pre-loaded Data
Recruiter Admin	`/login`	`recruiter@hirex.dev`	`Password123!`	4 Open Jobs, 3 Applicants in Pipeline
Interviewer	`/login`	`rahul.sharma@hirex.dev`	`Password123!`	1 Confirmed Interview, 1 Assigned Interview
Candidate	`/login`	`rahul@hirex.dev`	`Password123!`	1 Applied Job, 1 Confirmed Interview, Resume

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4. Browser, Device & Responsiveness Matrix

Device / Platform	Viewport	Target Browsers	Testing Focus
Desktop (Primary)	1920x1080, 1440x900	Chrome (latest), Edge (latest), Firefox (latest), Safari (latest)	Full layout, Kanban drag-and-drop, scorecards, document previews
Tablet	768x1024, 1024x1366	iPad Safari, Android Chrome	Responsive sidebars, table scroll, stepper navigation
Mobile (Candidate)	375x812, 412x915	Mobile Chrome, Mobile Safari	Job discovery, dynamic application wizard horizontal scroll, slot selection

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5. Priority & Severity Classifications

Priority Definitions

- **P0 (Critical / Blocker):** Core candidate application, slot booking, login, or stage progression is completely non-functional with no workaround.
- **P1 (High):** Major feature failure (e.g., offer generation fails, scorecard update throws error, document upload fails).
- **P2 (Medium):** Non-blocking issue with functional workaround (e.g., UI alignment issue, filter dropdown glitch, secondary notification delayed).
- **P3 (Low):** Minor cosmetic, typography, or animation imperfection.

Severity Definitions

- **S1 (Blocker):** Data corruption, crash, security vulnerability, cross-tenant data leak.
- **S2 (Major):** Business workflow failure without viable workaround.
- **S3 (Minor):** Annoying UI bug or edge-case input failure that does not halt the hiring flow.
- **S4 (Trivial):** Spelling mistake, color shade mismatch, or minor visual padding glitch.

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6. Entry & Exit Criteria

6.1 Entry Criteria for Testing

1. Database schema migration (`20261023_prd_gap_closure`) successfully applied.
2. `npx tsc --noEmit` passes with 0 errors across all 4 project folders.
3. Automated unit test suite passes with 100% pass rate.
4. Test accounts and seed data are populated via `npx prisma db seed`.

6.2 Exit Criteria (Production Readiness)

1. 100% of P0 and P1 test cases executed with **PASS** status.
2. 0 open S1 or S2 security, data integrity, or crash defects.
3. State machine validation verified: zero unauthorized reversions to `APPLIED`.
4. Tenant isolation verified: zero cross-tenant record leakage between companies.